

ECMM461

UNIVERSITY OF EXETER

**COLLEGE OF ENGINEERING, MATHEMATICS
AND PHYSICAL SCIENCES**

COMPUTER SCIENCE

Examination, May 2021

High Performance Computing

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Duration: TWO HOURS + 30 MINUTES UPLOAD TIME

Answer ALL questions.

The marks for this module are calculated from 50% of the percentage mark for this paper plus 50% of the percentage mark for associated coursework.

This is an OPEN BOOK examination.

Question 1

- (a) The graph below plots processor clock frequency against time for different processors produced since 1985.

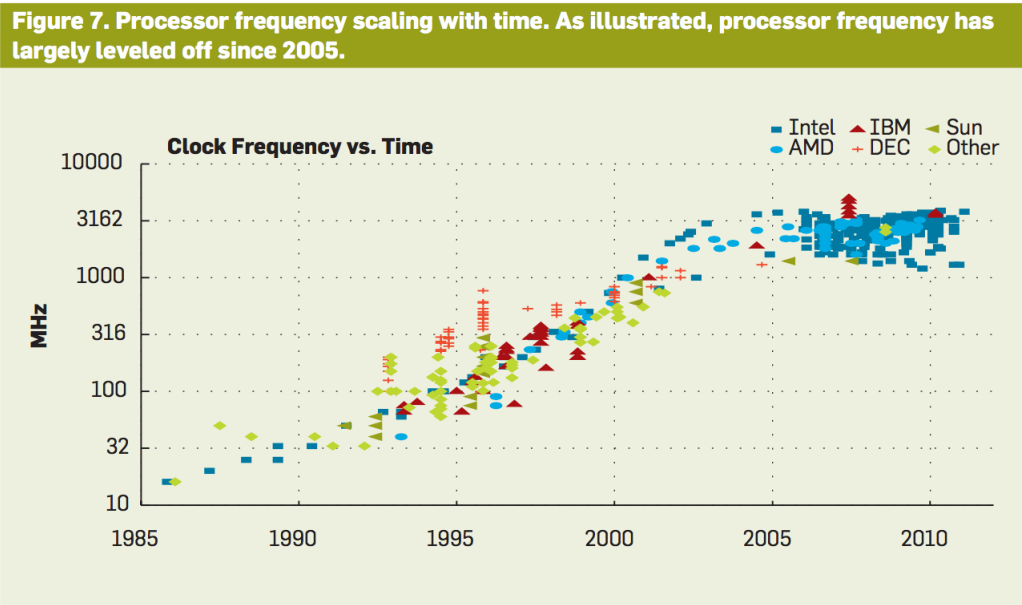


Figure 1: This figure shows changes in processor clock frequency from 1985 up to the mid-2010's. Time is plotted on the horizontal axis using a linear scale. Processor clock frequency in MHz is plotted on the vertical axis using a logarithmic scale. Processor clock frequency increases approximately linearly from 1985 until the mid-2000's. After the mid-2000's typical processor clock frequencies do not increase.

- (i) What enabled processor clock frequencies to increase prior to 2005?
(2 marks)
- (ii) Why are processor clock frequencies no longer increasing?
(2 marks)
- (iii) What are the implications for high performance computing?
(3 marks)

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- (b) The fastest computer on the June 2011 Top500 list was called the K computer. The following table shows information about the K Computer taken from the Top500 website.

Manufacturer:	Fujitsu
Cores:	705,024
Processor:	SPARC64 VIIIfx 8C 2GHz
Interconnect:	Custom Interconnect
Linpack Performance (Rmax):	10,510 TFlop/s
Theoretical Peak (Rpeak)	11,280.4 TFlop/s
HPCG performance:	602.736 TFlop/s
Power consumption:	12,659.89 kW
Operating system:	Linux

- (i) What is the LINPACK efficiency of the K computer? Show your working.

(2 marks)

- (ii) Why is the HPCG performance figure significantly less than the Linpack performance figure?

(2 marks)

- (iii) Based on the information shown above would you class the K computer as a commodity cluster or an MPP (Massively Parallel Processor) and why?

(2 marks)

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- (c) The following C program calculates a Gaussian probability density distribution and converts it to a cumulative distribution. The constant `M_PI` is defined in the `math.h` header file.

```
#include <stdio.h>
#include <math.h>

int main(){

    const int N=201;          // Number of points
    const float dx=0.5;       // Spacing of points
    const float x0=50.0;      // Centre of Gaussian
    const float sigma=10.0;   // Width of Gaussian
    float a[N];               // Gaussian distribution
    float normConst;          // Normalisation
    float x;                  // Argument of function
    float z;                  // Normalised argument

    // Calculate the normalisation constant
    normConst = 1.0 / ( sigma * sqrt(2.0*M_PI) );

    // Loop 1: Calculate Gaussian probability
    for (int i=0; i<N; i++){
        x = ( (float) i ) * dx;
        z = (x-x0) / sigma;
        a[i] = normConst * exp (-0.5*z*z);
    }
    printf("Maximum x value: %g\n", x);

    // Loop 2: Convert to a cumulative distribution
    for (int i=1; i<N; i++){
        a[i] = a[i-1] + a[i]*dx;
    }

    return 0;
}
```

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- (i) Consider whether the first loop (labelled “Loop 1”) contains a loop-carried dependency. If there is a dependency say what type of dependency it is and why it exists. If there is no dependency explain why no dependency exists.

(3 marks)

- (ii) Consider whether the first loop can be parallelised using OpenMP. If the loop can be parallelised then show the directive or directives you would use. You should ensure that variables are explicitly scoped unless they are declared with the `const` qualifier. If the loop cannot be parallelised by adding one or more OpenMP directives then say that the loop cannot be parallelised and explain why.

(3 marks)

- (iii) Consider whether the second loop (labelled “Loop 2”) contains a loop-carried dependency. If there is a dependency say what type of dependency it is and why it exists. If there is no dependency explain why no dependency exists.

(3 marks)

- (iv) Consider whether the second loop can be parallelised using OpenMP. If the loop can be parallelised then show the directive or directives you would use. You should ensure that variables are explicitly scoped unless they are declared with the `const` qualifier. If the loop cannot be parallelised by adding one or more OpenMP directives then say that the loop cannot be parallelised and explain why.

(3 marks)

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- (d) The following program calculates values of the function $y(x) = \sin(nx)$ for $n = 1, 2, 3, 4$. The constant `M_PI` is defined in the `math.h` header file.

```
#include <stdio.h>
#include <math.h>

int main(){

    const int NPOINTS=1001;
    const int NCURVES=4;
    double dtheta;
    double y[NCURVES][NPOINTS];
    double theta[NPOINTS];

    dtheta = 2*M_PI / ( (float) (NPOINTS-1) );

    for (int n=0; n<NCURVES; n++){
        for(int i=0; i<NPOINTS; i++){
            theta[i] = ( (float) i) * dtheta;
            y[n][i] = sin( ((float) (n+1)) * theta[i]);
        }
    }

    return 0;
}
```

- (i) Show how you would parallelise this program using OpenMP ensuring that it can scale to make effective use of a modern multi-core processor.

(4 marks)

- (ii) Explain why you chose to parallelise the program in this way.

(3 marks)

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- (e) MPI is a widely used parallel programming technology which can be used to write parallel programs to run on a wide range of architectures. OpenMP is another parallel programming technology which can be used instead of, or in combination with, MPI.
- (i) Describe four situations where you would choose to use OpenMP instead of, or in combination with, MPI.

(8 marks)

(Total 40 marks)

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Question 2

- (a) The following paragraph describes the Archer HPC system:

“The ARCHER hardware consists of the Cray XC30 MPP supercomputer, external login nodes and post processing nodes, and the associated filesystems. There are 4920 compute nodes in ARCHER phase 2 and each compute node has two 12-core Intel Ivy Bridge series processors giving a total of 118,080 processing cores. Each node has a total of 64 GB of memory with a subset of large memory nodes having 128 GB.”

- (i) What is the largest number of processor cores available to an OpenMP program running on Archer and why?

(2 marks)

- (ii) What is the largest number of processor cores available to an MPI program running on Archer and why?

(2 marks)

- (iii) The SHMEM library can be used to write parallel programs. According to Wikipedia:

“SHMEM (from Cray Research’s “shared memory” library) is a family of parallel programming libraries, providing one-sided, RDMA, parallel-processing interfaces for low-latency distributed-memory supercomputers. The SHMEM acronym was subsequently reverse engineered to mean “Symmetric Hierarchical MEMory”. Later it was expanded to distributed memory parallel computer clusters, and is used as parallel programming interface or as low-level interface to build partitioned global address space (PGAS) systems and languages.”

What is the largest number of processor cores available to a program using SHMEM running on Archer and why?

(2 marks)

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(b) Performing large scale calculations requires the use of parallel programs which can scale effectively to use many processors. Understanding the efficiency and scaling properties of a parallel program is important in order to make effective use of HPC systems.

(i) If a program takes 120 minutes to run on one processor and the maximum parallel speed up is 5 what is the shortest possible run time for the program if the total problem size remains fixed? Show your working.

(3 marks)

(ii) If the serial fraction of a program is $s = 0.1$ how many processors are required to give a speed up of 5 according to idealised strong scaling?

(5 marks)

(iii) If the serial fraction of a program is $s = 0.1$ what is the smallest number of processors required to give a speed up of at least 5 according to idealised weak scaling?

(5 marks)

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- (c) Consider a two-dimensional partial differential equation (PDE) solver which solves a PDE with one unknown variable u . To solve the PDE it is necessary to calculate values for the second derivative

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

This expression is evaluated using the finite difference stencil

$$\text{del2u} = (u(i+1, j) - 2*u(i, j) + u(i-1, j)) / (dx*dx) + (u(i, j+1) - 2*u(i, j) + u(i, j-1)) / (dy*dy)$$

The calculation is domain decomposed such that each sub-domain comprises 1000×1000 grid points. In addition to these grid points each sub-domain stores halo values which are copies of the data on neighbouring processors.

- (i) How many values need to be exchanged when one halo is sent to one neighbouring process?

(1 mark)

- (ii) How many bytes need to be sent if the values are represented using IEEE754 double precision?

(2 marks)

- (iii) What is the transmission time for sending a single halo over a network with a latency of $2\mu\text{s}$ (2×10^{-6} s) and a bandwidth of 25 GB/s (25 gigabytes per second)? Show your working.

(3 marks)

- (iv) Is the bandwidth or the latency more important for determining the transmission time for sending one halo? Explain your reasoning.

(2 marks)

- (v) The PDE solver is modified so that it solves a three-dimensional PDE and each sub-domain contains $1000 \times 1000 \times 1000$ grid points. Is the bandwidth or the latency more important for determining the transmission time for sending one halo in the modified program? Explain your reasoning.

(3 marks)

(Total 30 marks)

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Question 3

- (a) A cluster comprises 64 compute nodes with each compute node constructed from a two socket motherboard. Each socket has a multi-core processor comprising 16 cores with a clock frequency of 2.6 GHz. The instruction set supported by the processors is capable of delivering 4 floating point operations per cycle.

- (i) What is the peak performance of a single processor core? Show your working.

(2 marks)

- (ii) What is the peak performance of a single compute node? Show your working.

(2 marks)

- (iii) What is the peak performance of the whole cluster? Show your working.

(2 marks)

- (iv) You run a matrix solver based on the BLAS routine DGEMM on one compute node. Do you expect it to achieve large fraction of the compute node peak performance? Explain your reasoning.

(3 marks)

- (v) You run a sparse matrix solver on one compute node. Do you expect it to achieve a large fraction of the compute node peak performance? Explain your reasoning.

(3 marks)

- (b) The following three arrays represent a sparse matrix in compressed sparse row format. The arrays represent the values (`vals`), column indices (`col_ind`) and row pointers (`row_ptr`) with numbering starting from one.

vals	3	1	9	13	1	8	10	row_ptr	1	2	5	6	8
col_ind	2	1	2	4	4	2	4						

Write out the matrix in full as a grid of numbers including the zeros.

(9 marks)

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- (c) A finite difference method is used to solve the advection equation. The time step used to compute the solution is calculated according to the constraint

$$\Delta t = C \frac{\Delta x}{v_{\max}}$$

where Δt is the time step, Δx is the spacing between grid points, $v_{\max}$ is the maximum velocity in the calculation and C is a constant with a value between 0 and 1.

- (i) How does the computational cost of calculating a one-dimensional solution with N grid points compare with the computational cost of calculating a three-dimensional solution with N grid points in each dimension? You should assume that spatial resolution and number of time steps calculated are the same in both cases. Explain your reasoning.

(2 marks)

- (ii) How does the computational cost of the three-dimensional calculation increase if Δx is reduced by a factor of 10? You should assume that end time of the calculation and the extent of the computational domain remain constant. Explain your reasoning.

(4 marks)

- (iii) Consider a different solver which calculates solutions to the Navier-Stokes equations describing fluid dynamics. With this solver the maximum velocity present in the solution increases when the spatial resolution is increased by adding more grid points. Does this make the computational cost larger or smaller compared to the case where the maximum velocity does not change? You should assume that the extent of the computational domain and the number of time steps calculated remain constant. Explain your reasoning.

(3 marks)

(Total 30 marks)